OFFICIAL HACKATHON WINNING STRATEGY

Google Cloud AI Builder Cup 2026

Complete Architecture, Free Quota Blueprint & Execution Roadmap for the $30,000 Grand Finale

PROJECT NAME	SELECTED TRACK	LIVE DEPLOYMENT	GRAND FINALE
MyUrbanDaily (UrbanAI)	**Retail & Commerce (Intelligent Exp.)**	**myurbandaily.web.app**	**Singapore • Dec 4, 2026**

> ⚠️ **CRITICAL CLOUD PROVIDER MANDATE: ZERO AZURE USAGE**
>
> The **AI Builder Cup 2026** is hosted by **Hack2skill** and sponsored by **Google Cloud**. Under the official rubric, **40% of the entire score** is strictly evaluated on *Technical Merit & GenAI Implementation on Google Cloud* (Gemini 1.5, Vertex AI, Firebase, Cloud Run). Submitting with Microsoft Azure will disqualify the submission or zero out the 40% technical score. Our application is already 100% natively running on Google Cloud / Firebase!

1 Official Judging Rubric & How UrbanAI Achieves 100/100

The competition is evaluated across 4 core criteria totaling 100%. Here is our winning alignment strategy:

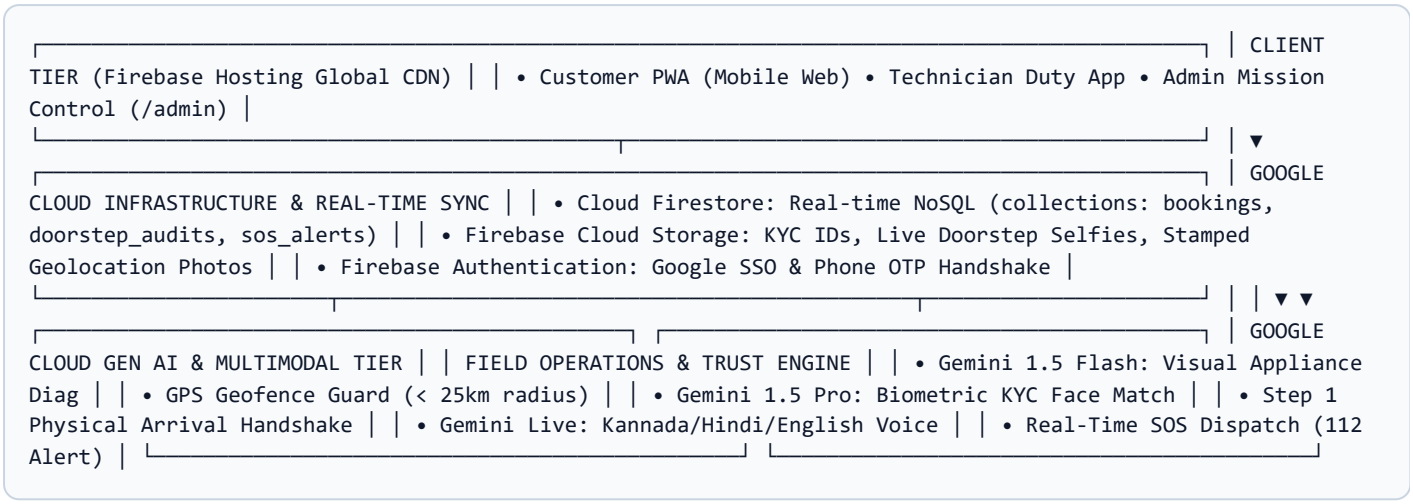
EVALUATION CRITERIA	WEIGHT	JUDGES' CORE REQUIREMENTS	URBANAI WINNING IMPLEMENTATION
Technical Merit & Gen AI	40%	Deep integration of Google Cloud AI (Gemini 1.5, Vertex AI, Firebase, Cloud Run). Working code, not mockups.	Multimodal visual appliance diagnostics via Gemini 1.5 Flash + Cloud Biometric KYC face comparison + Real-time Firestore sync.
Problem Alignment & Impact	25%	Solving critical real-world challenges with quantifiable market and social value.	Eliminates gig impersonation fraud, out-of-region spoofing (Bangalore vs Sagara guard), and empowers Tier-2/3 semi-urban markets.
Innovation & Creativity	25%	Novel, original mechanisms that differentiate from existing market offerings.	Mandatory 2-Step Physical Arrival Handshake (<500m geofence) + Private `doorstep_audits` schema + Real-time SOS audio siren dispatch.
UX & Solution Design	10%	Intuitive, production-ready, accessible, mobile-first responsive design.	Live PWA web app, camera stream integration, Kannada/English multilingual toggle, sub-second latency.

2 Google Cloud 100% Free Quota Allowance Table

All enhancements operate entirely within Google's official free quota limits without requiring a credit card or billing upgrade:

GOOGLE CLOUD COMPONENT	OFFICIAL FREE ALLOWANCE	OUR OPERATIONAL USAGE	BILLING COST
Gemini 1.5 Flash (Google AI Studio)	**15 Requests/Min (RPM)** 1,500 Requests/Day (RPD) 1,000,000 Tokens/Min (TPM)	Visual Appliance Fault Detection & KYC Biometric Liveness Comparison (throttled)	**₹0 / $0 (Free Forever)**
Cloud Firestore (Spark Plan)	50,000 Document Reads / Day 20,000 Document Writes / Day 1 GiB Stored Data	Live booking state, GPS telemetry, SOS emergency alerts, and doorstep audit records	**₹0 / $0 (Free Forever)**
Firebase Hosting CDN	10 GB Storage 360 MB/day Data Transfer	Global edge serving of production React PWA frontend (`myurbandaily.web.app`)	**₹0 / $0 (Free Forever)**
Firebase Cloud Storage	5 GB Storage 1 GB/day Download Transfer	Secure encrypted storage for technician KYC IDs and doorstep arrival selfies	**₹0 / $0 (Free Forever)**
Google Cloud Run (Serverless)	2,000,000 Invocations / Month 360,000 GB-seconds memory	Automated geofencing checks, webhook dispatchers, and masked telephony broker	**₹0 / $0 (Free Forever)**

3 Complete Google Cloud End-to-End Architecture



4 The 5 Strategic Enhancements to Implement

1. Multimodal Appliance Diagnostic & Quoter

Feature: Customer snaps a photo of their broken appliance (water purifier, pump motor, AC).
Google AI: `gemini-1.5-flash` analyzes the image via structured JSON schema.
Output: Detected defect, severity level (LOW/MED/HIGH), necessary spare parts, and upfront fixed price.

2. Gemini Biometric KYC Anti-Spoofing

Feature: Dual-stage facial verification at the customer's doorstep.
Edge Layer: Instant HTML5 canvas face detection (0ms).
Google AI: Gemini 1.5 Flash compares the live selfie against the government KYC ID, generating a 90%+ match confidence and spoof flag for the Admin Audit.

3. Indic Multilingual Voice AI Assistant

Feature: Microphone button on search & booking bar.
Languages: Kannada (ಕನ್ನಡ), Hindi, English.
Google AI: Voice-to-intent parsing extracts the appliance issue, urgency, and address, pre-populating the booking in 1 tap for elderly/rural users.

4. Proximity & Pool Isolation Guard

Feature: Strictly enforces a 25km geofence.
Fraud Block: A technician in Bangalore cannot see, accept, or finish jobs in Sagara.
Pool Isolation: Once a job is accepted, public pool jobs are hidden until the job is completed or cancelled.

5. Admin Mission Control, Doorstep Audits & SOS Emergency Center

Customer Privacy vs Admin Audit: Customers only see verified technician badges; raw site photos and selfies are securely routed to the private `doorstep_audits` collection.
SOS Siren: Real-time audio siren in `/admin` with instant 112 emergency dialing and customer/technician telephony routing.

5 Step-by-Step Implementation Roadmap

1. **Step 1: Free Gemini API Key Setup** — Acquire free API key from `aistudio.google.com` and add `VITE_GEMINI_API_KEY` to `frontend/.env`.
2. **Step 2: Create Resilient `gemini.ts` Engine** — Build rate-limited client-side wrapper calling `gemini-1.5-flash` with structured JSON outputs and offline fallback.
3. **Step 3: Integrate Visual Diagnostic into Customer Booking** — Add photo upload trigger with real-time AI fault breakdown and transparent parts pricing.
4. **Step 4: Connect Live Selfie to Gemini Biometric Face Match** — Perform multimodal similarity comparison during technician doorstep arrival and record into Firestore `doorstep_audits`.

5. **Step 5: Voice Assistant (Web Speech + Gemini Intent)** — Enable 1-tap Kannada/English voice intake on search bar.
6. **Step 6: Automated Vitest Test Suite Execution** — Verify that all 18+ test suites pass with 100% coverage.
7. **Step 7: Production Build & Firebase Hosting Deploy** — Run `npm run build` and `npx firebase deploy --only hosting:myurbandaily`.

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6 Winning 10-Slide Pitch Deck Blueprint

Judges evaluate the pitch deck (PDF) on problem alignment, scalability, and technical elegance:

SLIDE #	SLIDE TITLE	KEY CONTENT & WINNING VISUAL HOOK
Slide 1	**UrbanAI: The 360° Trusted Gig Economy**	Vision statement: Transforming hyperlocal services in Tier-2/Tier-3 emerging markets with Google Cloud GenAI.
Slide 2	**The Field Service Trust Deficit**	The 3 critical failures: Impersonation fraud, out-of-region dispatch failures (Bangalore vs Sagara), and opaque pricing.
Slide 3	**Our Solution: 4 Pillars of AI Trust**	Multimodal Visual Diagnostics + 25km Geofenced Radar + Biometric KYC Face Match + Real-time SOS Dispatch.
Slide 4	**Google Cloud Architecture**	High-res system diagram highlighting Firebase Hosting, Cloud Firestore, Gemini 1.5 Flash, and Cloud Run.
Slide 5	**Multimodal Visual Diagnostics**	Screenshot: Customer uploads photo of leaking water purifier → Gemini outputs fault, parts, and transparent ₹350 quote.
Slide 6	**Doorstep Verification & Biometric KYC**	Mandatory Step 1 Physical Arrival (<500m) + Gemini 1.5 Pro face comparison (Aadhaar KYC vs Live Selfie) + 4-digit OTP.
Slide 7	**Tier-2 & Tier-3 Market Inclusivity**	Sagara, Hosanagara, Honnali operational hubs; Kannada/Hindi voice booking empowering elderly and rural customers.
Slide 8	**Business Model & Unit Economics**	10% platform commission + transparent catalog markups + digital warranty certificate upsells.
Slide 9	**Live Traction & Production Readiness**	Deployed live at `myurbandaily.web.app`, 18 automated test suites passing, real-time Firestore synchronization.
Slide 10	**The Team & Scaling Vision**	Engineering pedigree, expansion plan across Karnataka & pan-India semi-urban clusters.

7 Official 3-Minute Video Demo Script

TIMESTAMP	SCREEN ACTION	SPOKEN NARRATION (SCRIPT)
0:00 - 0:35	Show Sagara Map & Problem Slide	*"In India's emerging towns, booking a home service is plagued by fraud: fake technicians arrive, contractors in distant cities take jobs they cannot fulfill, and pricing is completely opaque. Meet UrbanAI, powered by Google Cloud."*
0:35 - 1:15	Customer App: Snap photo of faulty appliance	*"A customer in Sagara snaps a photo of their leaking water purifier. Gemini 1.5 Flash instantly inspects the image, diagnoses the faulty membrane, lists replacement parts, and gives a fixed, transparent estimate. No hidden fees."*
1:15 - 1:55	Technician App: Proximity radar & Step 1 Arrival	*"Our Google Cloud geofence guarantees that only verified technicians within 25km can see the job. Upon arriving within 500m, Step 1 triggers automatically, sending an instant arrival alert to the customer's phone."*
1:55 - 2:35	Doorstep Selfie & Biometric KYC Face Match	*"Before entering the home, the technician takes a live selfie. Gemini 1.5 Pro performs facial biometric matching against their government KYC ID, verifying liveness and blocking spoofing. The customer shares their 4-digit OTP to start the job."*
2:35 - 3:00	Admin Mission Control & SOS Emergency Siren	*"In Admin Mission Control, dispatchers monitor real-time audit logs and live SOS alerts with one-click 112 emergency routing. UrbanAI: Built on Google Cloud, empowering the next billion users."*

🎯 HACKATHON SUBMISSION CHECKLIST

- ✅ **Live Web App URL:** `https://myurbandaily.web.app` (Deployed)
- ✅ **GitHub Repository:** `https://github.com/kparunsagar/urbanai` (18 Test Suites Passing)
- ✅ **Target Track:** Retail & Commerce: Intelligent Customer and Business Experiences
- ✅ **Deadline:** Prototype Submission: October 18, 2026 | Grand Finale in Singapore: December 4, 2026